Noman Bin Morshed

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Education

University of Dhaka, Bangladesh 2022–2026

Bachelor of Science (Honours) in Nuclear Engineering **CGPA: 3.73/4.00**

Undergraduate Thesis: A Quantum-Inspired Hybrid Framework for Nuclear Fuel Loading Optimization. Developed a hybrid framework integrating QUBO-based simulated annealing, simulated quantum annealing, CNN surrogate modeling, and genetic algorithms for nuclear fuel loading optimization, with OpenMC-based verification.

Relevant coursework: Nuclear Physics, Quantum Mechanics, Nuclear Fusion, Nuclear Instrumentation, Numerical Methods, Statistics, Computer Programming (C++).

Research Experience

PWR Loading Pattern Optimization Using a Convolutional Neural Network Surrogate with Conformal Safety Calibration and Annealing-Initialized Genetic Search

- Developed an active-learning CNN surrogate for static 2D, 193-assembly PWR loading optimization, combining targeted OpenMC evaluations, QUBO annealing warm starts, and constraint-preserving genetic search.
- Integrated strict out-of-fold residual modeling with one-sided conformal safety calibration; achieved 96.39% joint coverage on 360 prospective OpenMC evaluations, with all selected patterns satisfying the study's physical constraints.
- Conducted controlled statistical comparisons identifying local memetic refinement as the main driver of objective improvement; effects of global search strategy and simulated versus simulated quantum annealing initialization were small.

Neutron-Damage-Dependent Optimization of Thermal Detritiation in Actively Cooled W/Cu/CuCrZr Monoblocks

- Investigated neutron-damage effects on tritium retention, trap evolution, and thermal release using coupled transport and trapping simulations in FESTIM.
- Optimized bake temperature and duration across damage levels to maximize tritium removal while assessing coolant-transfer trade-offs, using an inventory-balanced operating map of residual, vacuum-release, and coolant-transfer fractions.

Selected Research Projects

QWorld — QIntern 2026 Research Internship

Learning-Guided Graph Coarsening for the Capacitated Vehicle Routing Problem

- Developed a PyTorch graph neural network to guide customer aggregation, reducing routing problem size while preserving vehicle-capacity constraints.
- Integrated classical and QUBO-based solvers with route reconstruction and cross-cluster variable neighborhood search; evaluated 14 benchmark instances using source-held-out validation and controlled ablations.
- Achieved a mean 4.38% route-cost reduction through cross-cluster refinement of GNN-coarsened solutions; all 42 runs across three random seeds satisfied capacity and customer-coverage constraints.

WISER Summer Program 2025 — Official Industry Track Project

Quantum Galton Board Simulator

- Successfully completed the assessed industry-track project and earned official WISER project-track certification.
- Implemented normal, exponential, and Hadamard quantum-walk distributions in Qiskit using the Carney–Varcoe Universal Statistical Simulator framework.
- Designed ancilla-assisted, CSWAP-based circuits and evaluated fidelity under IBM hardware noise models using standard and AI-assisted transpilation.

Experience

IBM Quantum — Qiskit Advocate *Remote*

- Completed the Qiskit Advocate Training Program and contributed to community-led quantum simulation and circuit optimization projects.

Bangladesh Atomic Energy Commission *Industrial Training*

- Received hands-on training in nondestructive evaluation and nuclear material characterization using eddy current, magnetic particle, radiographic, and ultrasonic testing.

Technical Skills

Programming	Python, MATLAB, C++
Scientific Computing	FESTIM, OpenMC, Quantum ESPRESSO, NumPy, SciPy, Pandas, NetworkX
Machine Learning	PyTorch, TensorFlow
Optimization Libraries	PyQUBO, PyGAD, OpenJij
Quantum Computing	Qiskit, PennyLane, D-Wave Ocean SDK
Development Tools	Git, Linux
Engineering Software	ANSYS HFSS, KiCad

Selected Academic Programs

Intro to Plasma and Fusion (2026)	Princeton Plasma Physics Laboratory (PPPL) and Oak Ridge National Laboratory (ORNL). Topics: Plasma Physics, Magnetic Fusion, Fusion Materials, AI for Fusion.
MTV Summer School (2025)	University of Michigan. Topics: Radiation Detection and Imaging, Scintillator Synthesis, Monte Carlo Simulation Methods.
Quantum Program (2025)	Washington Institute for STEM, Entrepreneurship and Research. Topics: Quantum Linear Solvers, Quantum Algorithms for PDEs, Quantum Simulation, Lie Theory.

Selected Certifications

QSilver36, QCobalt, and QBronze–Qiskit (QWorld); Fundamentals of Quantum Algorithms (IBM); Trotterization and LCU Challenges (PennyLane); QML Workshop (Fundacja Quantum AI).